

Exploring Cluster Structures through Interactive Attribute Weighting

A Visual Analytics Interface for Clustering Evaluation

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1 Introduction

The project focuses on the development of a *Visual Analytics* interface designed to support the analysis, monitoring, and evaluation of data clustering. The main goal of the application is to provide the user with an interactive tool for exploring user-selected datasets, observing the structure of the data, and comparing different clustering approaches.

In particular, the interface allows the user to load or select a dataset and analyze its distribution through different visualizations. These visualizations were designed to make cluster interpretation more immediate, highlight possible patterns in the data, and facilitate the comparison of results obtained through different approaches.

The system is organized around two main types of visualization: the *Cluster View* and the *Interactive View*. The first is dedicated to representing clustering results through specific charts, while the second is intended to provide a more dynamic and interactive exploration of the data.

The most important motivation behind the project is the need to understand which attributes have the greatest influence on the clustering process. The tool was created not to evaluate the final clusterization, but to investigate how specific attributes may affect the formation of clusters and, in some cases, lead to incorrect assignments. This includes the possibility of identifying attributes that contribute to false positives or false

negatives, where data points are grouped together or separated in a way that does not reflect the expected structure of the dataset.

2 Project Objectives

The main objectives of the project can be summarized as follows:

- provide an intuitive interface for analyzing user-selected datasets;
- enable the visualization of data clustering through different charts;
- allow the comparison between automatically computed clustering and clustering based on the labels available in the dataset;
- support the user in evaluating the quality and consistency of the identified clusters;
- help the user understand the importance of individual attributes in the clustering process;
- identify attributes that may negatively influence the clusterization and cause false positive or false negative assignments;
- make data exploration clearer through visual and interactive tools.

3 General Description of the Interface

The interface was designed to help the user better understand the internal structure of the data. After selecting a dataset, the user can access different views that display the data from different perspectives. This approach makes it possible to move from a general view of the distribution of points to a more specific evaluation of the clusters. The default datasets that we provide are 'wine.csv' and 'iris.csv', whereas the user can easily upload any dataset that they desire. For each uploaded dataset, the application initially sets the first attribute as the default label or reference attribute. However, the user can change this selection and decide which attribute should be used as the label for the clustering analysis. In this way, the clustering process is not fixed to a predefined class, but can be adapted according to the attributes that the user considers most relevant.

The separation between the *Cluster View* and the *Interactive View* makes it possible to distinguish between visualizations mainly focused on clustering results and visualizations focused on direct exploration. In this way, the user can both observe the results produced by the algorithms and interact with the data to investigate specific aspects of interest.

4 Cluster View

The *Cluster View* is the section of the interface dedicated to visualizing the clusters present in the dataset. In this view, the user can observe different types of charts, which are useful for comparing alternative ways of grouping the data. At this stage, two main visualizations have been implemented and considered: a chart based on the K-Means algorithm and a clustering chart based on the labels of the dataset.

Figure 1 shows the initial view of the interface as it appears when the user first opens the application. This view provides the starting point for selecting or uploading a dataset and accessing the available clustering visualizations.

At the side of the four main visualization windows, the interface includes a *Weight Control Panel*. This panel lists the attributes of the selected dataset and excludes the attribute currently selected as the clustering or label reference. Since the user can change this reference attribute, the set of weighted attributes is updated accordingly. Through this panel, the user can modify the weight assigned to each available attribute and analyze how different weighting configurations affect the clustering result.

Inside the panel, there is also a switch button called *Continuous View*. When this option is activated, the user

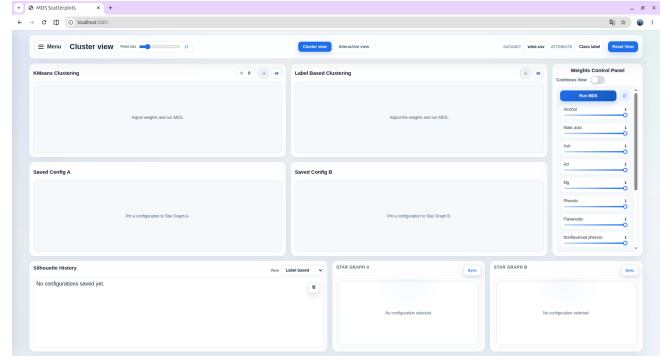


Figure 1: *Initial view of the interface when the application is opened.*

can see in real time how changing the weight of an attribute is reflected in the two graphs above. This makes the exploration more immediate, since each modification is directly propagated to the visualizations without requiring an additional action.

If *Continuous View* is not activated, the user can manually define a configuration by changing the weight of one or more attributes and then click the *Run MDS* button to plot the result on the graph. In this case, a point appears in the bottom-left graph, called *Silhouette History*, representing the specific configuration that has been generated and saved. A similar operation is also possible while using *Continuous View*: once the switch is activated, a *Save Configuration* button appears, allowing the user to store the current weighting configuration as a point in the same graph.

The *Silhouette History* graph is based on the silhouette coefficient, a metric used to evaluate the quality of a clustering result. The silhouette coefficient measures how similar a data point is to the other points in its own cluster compared to the points belonging to the nearest different cluster. A high silhouette value indicates that the point is well matched to its assigned cluster and clearly separated from other clusters, while a low or negative value suggests that the point may be close to another cluster or incorrectly assigned. For this reason, the silhouette coefficient is useful for comparing different weighting configurations and understanding whether changing the importance of some attributes improves or worsens the cluster separation.

Next to the *Silhouette History* graph, the interface also provides two star graphs that are used to inspect and compare saved weighting configurations. Each star graph represents one selected configuration of attribute weights: every axis corresponds to an attribute, and the shape of the graph shows how much importance has been assigned to each one. This representation makes it easier to understand the structure of a configuration and to compare how different attributes contribute to the clustering process.

Figure 2 shows an example in which multiple weight-

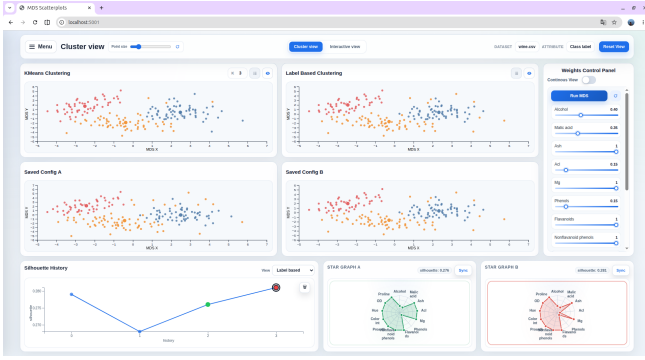


Figure 2: Example of saved weight configurations displayed through the *Silhouette History* graph and the two star graphs.

ing configurations have been saved through the interface. The *Silhouette History* graph displays the silhouette values associated with these configurations. By clicking on two different points in the *Silhouette History* graph, the user can select two saved configurations and display them in the two star graphs. This allows the user to directly compare two different arrangements of weights.

When the star graphs are plotted, the application simultaneously displays the corresponding clustering graphs in the two saved-configuration windows. Depending on the view selected from the drop-down menu in the *Silhouette History* graph, these can be either the label-based clustering graphs or the K-Means-based graphs. In this way, the user can compare not only the numerical and visual distribution of weights, but also the effect that each configuration has on the resulting clustering visualization. This comparison helps the user study how changes in attribute importance influence the separation of the data and whether certain weighting choices improve or worsen the cluster structure.

The *Sync* buttons associated with the star graphs allow the user to directly synchronize the *Weight Control Panel* with the weights represented by a selected star graph. By using this function, the user can quickly restore or reuse a saved configuration without manually adjusting each attribute weight again.

4.1 K-Means-Based Chart

One of the visualizations available in the *Cluster View* is the chart computed using the K-Means algorithm. This algorithm divides the data into a number k of clusters, attempting to group similar points together and separate points that belong to different groups.

An important aspect of this visualization is that the user can directly choose the value of k . This feature makes the chart particularly useful for exploring how the cluster structure changes as the requested number of groups varies. For example, low values of k may highlight broad macro-structures in the dataset, while higher values may reveal more detailed subdivisions.

The K-Means-based visualization therefore makes it possible to interactively evaluate the effect of the parameter k on the final clustering result. The user can compare different configurations and observe whether the resulting clusters are compact, well separated, or possibly overlapping.

4.2 Label Based Clustering Graph

The second visualization available in the *Cluster View* is the *Label Based Clustering Graph*. Unlike the K-Means-based chart, this representation uses the labels already present in the dataset to distinguish between groups of data.

This type of chart is useful when the dataset contains a categorical variable or a reference class that can be used as a label. In this case, the points are colored or grouped according to the selected label, allowing the user to observe the real or expected distribution of classes within the visualized space.

The comparison between the *Label Based Clustering* graph and the *K-Means-based* chart makes it possible to evaluate how consistent the automatically identified clusters are with the labels of the dataset. If the groups produced by K-Means are similar to the label distribution, this may indicate a good correspondence between the structure of the data and the known categories. Conversely, significant differences may highlight overlaps, outliers, or a more complex data structure.

5 Interactive View

The *Interactive View* is the part of the interface dedicated to the dynamic exploration of the data. Its general purpose is to allow the user to interact directly with the visualizations, select elements of interest, and analyze specific regions of the dataset in greater detail.

Figure 3 shows the initial layout of the *Interactive View*. In this view, the interface contains two main graph windows, both representing MDS projections of the selected dataset. The first graph shows the unweighted MDS projection, while the second graph shows the weighted MDS projection. The weighted MDS changes according to the values assigned in the *Weight Control Panel*, located on the right side of the interface. Therefore, when the user modifies the attribute weights, the second graph updates in real-time to represent the new weighted configuration of the data.

A key aspect of this visualization is the possibility of selecting points directly on the MDS plots through a lasso-like interaction tool. The user can draw a selection area around a group of points in one of the two graphs, and the selected points are then highlighted in both representations using the chosen color. This linked selection mechanism makes it easier to compare the same subset

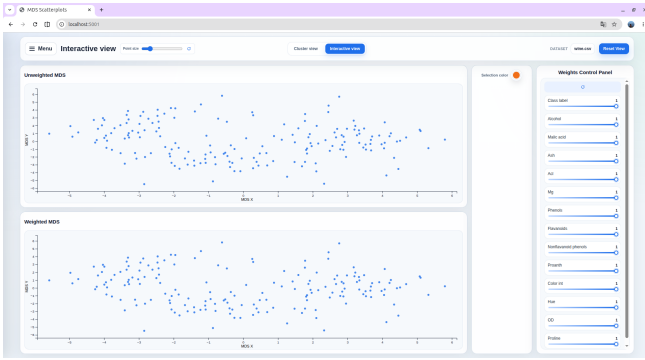


Figure 3: *Initial view of the Interactive View, with the unweighted MDS graph and the weighted MDS graph.*

of data across the unweighted and weighted MDS views, helping the user understand how the selected points change position when attribute weights are modified.

Figure 4 shows an example of this interaction. The lasso tool is used to select a subset of points in one MDS graph, and the same points are immediately highlighted in the other graph, allowing the user to visually track the selected data across both representations.

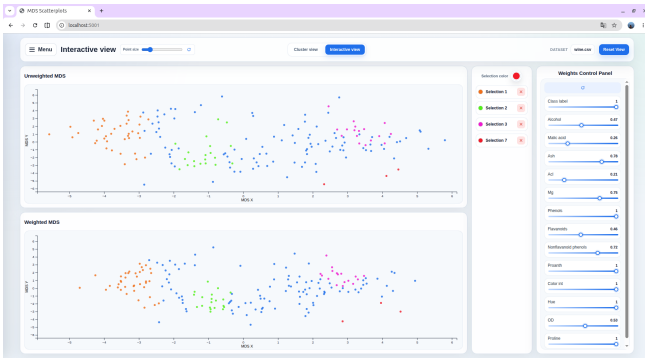


Figure 4: *Example of the lasso tool selecting points in one MDS graph and highlighting the same points in both Interactive View representations.*

The selection can also be deleted and the configuration can be restored by clicking on the refresh button. This view complements the *Cluster View*, since it allows the user to move from simply observing clusters to a more active and user-driven analysis.

6 User Insights and Practical Interpretation

The interface can provide several useful insights to users who want to understand how a dataset behaves under different clustering conditions. One of the most important insights concerns the role of individual attributes. By modifying the weights in the *Weight Control Panel*, the user can observe whether some attributes make the clusters more compact and separated, or whether they introduce noise and reduce the clarity of the clustering structure. When using the interface, the user should

first observe the initial unweighted configuration before changing any attribute weights. This baseline view is important because it provides a reference point against which all later configurations can be compared. Without this initial reference, it becomes more difficult to understand whether a change in the visualization is actually improving the clustering structure or simply transforming it in a different way.

A practical way to explore the dataset is to modify one attribute at a time. By increasing or decreasing a single weight and observing the changes in the visualizations, the user can isolate the contribution of that specific attribute. If the clusters become more separated and the silhouette value improves, the attribute is likely meaningful for the clustering structure. If the visualization becomes less clear or the silhouette value decreases, the attribute may introduce noise or distort the distances between points. Saved configurations should be used to compare different hypotheses. For example, the user can save one configuration where a specific attribute is emphasized and another configuration where the same attribute is reduced. The *Silhouette History* graph then makes it possible to compare these alternatives through their silhouette values, while the star graphs show how the corresponding weight distributions differ.

The user can also compare the K-means graph and the label based graph. If they show similar patterns, this suggests that the automatic grouping is consistent with the dataset labels. Otherwise, if they differ, the user should inspect whether the difference is caused by overlapping classes, outliers, an unsuitable value of k , or attributes that are not informative for the expected grouping.

The lasso tool in the *Interactive View* is useful for local analysis. Instead of only evaluating the dataset globally, the user can select a specific group of points and check how those points behave in both the unweighted and weighted MDS projections. This can reveal whether a group is stable across configurations, whether it becomes more separated after weighting, or whether it remains ambiguous regardless of the selected weights.

In practice, the most reliable interpretation comes from combining multiple signals: silhouette values, visual separation in the clustering graphs, star graph shapes, and point movement in the MDS projections. None of these elements should be considered in isolation; together, they help the user understand both the quality of a configuration and the reasons behind it.

7 Conclusions

This project developed a visual analytics interface for exploring clustering behavior in user-selected datasets. The system integrates clustering visualizations, attribute weighting, saved configurations, silhouette-based evaluation, and interactive point selection into a single analysis

workflow.

The main contribution of the interface is its ability to make the clustering process more interpretable. Rather than only showing a final clustering result, the tool allows the user to investigate how different attributes influence the organization of the data and how alternative weighting configurations affect cluster quality. The combination of the *Cluster View* and the *Interactive View* supports both global and local analysis. The former helps compare clustering configurations and evaluate their quality, while the latter allows the user to inspect specific subsets of points and observe how they behave under weighted and unweighted projections.

Overall, the interface provides a flexible environment for studying clustering results, identifying influential or problematic attributes, and comparing automatic clustering with known labels. By combining quantitative indicators with interactive visual exploration, the tool supports a more informed and interpretable analysis of complex datasets.